

Theta Closure in Modal Triplet Theory V: Weak-Angle Round Trip and the Non-Circularity Criterion

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Abstract

We reassess whether the weak mixing angle provides a redundant test of the selected MTT gauge profile. Let $r_{21} = I_2/I_1 = g_1^2/g_2^2$, with $g_1 = \sqrt{5/3} g_Y$. Then $\sin^2 \theta_W = 3r_{21}/(5 + 3r_{21})$ at the same scale and in the same scheme. For the selected SMDR profile at $Q = M_t$, this gives $\sin^2 \theta_W = 0.2346644 \pm 0.0000433$. Because r_{21} was itself obtained from the measured (g_Y, g_2) profile, this equality is an exact algebraic round trip, not a held-out prediction. Supplying an absolute g_2 from (G_F, m_W) does not remove that circularity. A genuine redundancy test requires an MTT source theorem selecting r_{21} without electroweak gauge data, followed by independent common-scheme transport. The obsolete one-loop 5 TeV calculation and its precision-prediction claim are withdrawn.

Revision note for this edition

Supersedes. *Theta Closure in Modal Triplet Theory V: Redundant Determination from Gauge Couplings and the Weak Mixing Angle*, first edition.

Reason. The weak angle was presented as an independent precision test even though it was algebraically reconstructed from the same measured gauge profile, and the scale used the retired few-TeV crossing.

Resolution. Version 2 derives the same-scheme round-trip formula, propagates its uncertainty at $Q = M_t$, and states an explicit non-circularity criterion for any future test.

Retained result. The weak angle remains an exact consistency identity for a supplied gauge-ratio profile.

Remaining boundary. A held-out test requires MTT to select r_{21} without electroweak gauge data and then transport it independently.

1 Selected inputs and scope

Paper I adopts the full-Standard-Model $\overline{\text{MS}}$ profile transported by SMDR v1.3 to

$$Q = M_t = 172.5590883453979 \text{ GeV.}$$

Its relevant entries are

$$g_Y = 0.3585945042 \pm 0.0000307251, \quad g_2 = 0.6475986708 \pm 0.0000287665,$$

or $g_1 = \sqrt{5/3} g_Y$ in GUT normalization. The associated overlap ratio is

$$r_{21} := \frac{I_2}{I_1} = 0.5110273 \pm 0.0001231.$$

Paper II realizes this ratio by calibrating its effective geometry. Paper III checks conditional compatibility of the quadratic norm. Neither paper currently supplies a value-source theorem selecting r_{21} independently of the gauge profile.

2 Same-scale weak-angle identity

The overlap convention is

$$\frac{1}{g_a^2} = \frac{I_a}{g_{10}^2}.$$

Therefore

$$r_{21} = \frac{I_2}{I_1} = \frac{g_1^2}{g_2^2}. \quad (2.1)$$

Using $g_Y^2 = (3/5)g_1^2$ and the $\overline{\text{MS}}$ definition

$$s_W^2(Q) := \frac{g_Y^2(Q)}{g_Y^2(Q) + g_2^2(Q)},$$

gives the exact identity

$$s_W^2(Q) = \frac{3r_{21}(Q)}{5 + 3r_{21}(Q)}. \quad (2.2)$$

Theorem 2.1 (Gauge-profile round-trip theorem). *If $r_{21}(Q)$ is constructed from the same measured common-scheme pair $(g_Y(Q), g_2(Q))$ through Equation (2.1), then evaluating Equation (2.2) returns the weak angle encoded in that pair. The equality is algebraic and cannot constitute an independent prediction.*

Proof. Substitution of $r_{21} = (5/3)g_Y^2/g_2^2$ into Equation (2.2) gives $g_Y^2/(g_Y^2 + g_2^2)$ identically. $\square$

For the selected ratio,

$$s_W^2(M_t) = 0.2346644 \pm 0.0000433, \quad (2.3)$$

where the uncertainty is propagated from the selected r_{21} row. Direct substitution of the selected g_Y and g_2 gives the same central value. This is a useful convention and arithmetic check, but it has no held-out status.

3 Why the (G_F, m_W) construction remains circular

An electroweak input such as (G_F, m_W) can set an absolute $SU(2)$ normalization, subject to radiative matching. It cannot make the weak-angle test independent when the ratio r_{21} still comes from the measured hypercharge-to- $SU(2)$ profile. Indeed, once g_2 and r_{21} are supplied, Equation (2.1) defines g_1 and hence the weak angle. The information being tested is already present in r_{21} .

The earlier tree-level value near 0.2312 used the obsolete 5 TeV profile and a one-loop return to M_Z . It is withdrawn as a prediction. Its threshold scan also showed that the apparent precision was not stable under the stated electroweak matching variation.

4 Criterion for a genuine redundant determination

A non-circular weak-angle test requires all of the following:

1. MTT geometry selects r_{21} without using g_Y , g_1 , g_2 , or $\sin^2 \theta_W$ as value inputs;

2. an absolute gauge normalization and matching scale are selected without using the held-out weak angle;
3. the boundary data are transported with a declared multi-loop scheme and threshold prescription; and
4. the resulting s_W^2 is compared with a measurement not used anywhere in selection, calibration, or branch choice.

Only then is Equation (2.2) a prediction map rather than a round-trip map. Current Papers I–III satisfy the algebraic and realization parts but not the first value-source condition.

5 Conclusion

The weak mixing angle is exactly consistent with the selected gauge profile, as it must be. The precise same-scale result is $s_W^2(M_t) = 0.2346644 \pm 0.0000433$. This validates the hypercharge normalization and overlap-ratio bookkeeping.

It does not add an independent Standard Model observable to MTT closure. The former non-circularity claim fails because the overlap ratio already contains the measured weak-angle information. The next theorem target is sharply defined: select r_{21} from MTT source geometry before consulting the electroweak gauge profile, then execute a held-out common-scheme comparison.

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